

Problem Statement:

This capstone project aims to develop an advanced indoor drone navigation system using the QAV250 platform and Simulink. The project will address the following objectives:

1. **Configure the QAV250 for indoor flight using the PX4 open-source stack (QGC).**
 - Establish various flight control modes (e.g., position/altitude control) to ensure stable and precise maneuvering within indoor environments. This involves setting up the PX4 firmware and tuning the parameters to achieve optimal performance.
2. **Configure the QAV250 for indoor flight using the PX4 Support Package within Simulink and utilize host target and SIH workflow.**
 - Leverage the PX4 Support Package in Simulink to create a seamless integration between the drone hardware and the simulation environment. This allows for real-time testing and validation of control algorithms using the host-target and Software-In-the-Loop (SIH) workflow.
3. **Develop a Simulink plant model for drone simulation.**
 - Create a plant model in Simulink that accurately represents the dynamics of the drone. The model should be modular, allowing it to be adapted for different PX4-supported multirotor frames. This flexibility ensures that the simulation can be used for various drone configurations without significant modifications.
4. **Design and implement a custom flight controller using Simulink.**
 - Develop a flight controller in Simulink that is modular and compatible with PX4-supported multirotor frames. This involves designing control algorithms that can be easily adapted to different drone platforms.
 - Explore and fine-tune examples from MathWorks Documentation to tailor the controller for optimal indoor flight performance. This step ensures that the controller is robust and capable of handling the unique challenges of indoor navigation.
 - Investigate and implement adaptive gain scheduling for the multirotor UAV, including PID auto-tuning. Adaptive gain scheduling allows the controller to adjust its parameters in real-time, improving stability and performance under varying flight conditions.
5. **Incorporate and validate sensor measurements from a camera and range sensors (e.g., LiDAR).**
 - Develop a pipeline for the Simulink PX4 flight controller to access and leverage external sensors such as cameras and LiDAR. These sensors provide critical data for navigation and obstacle detection, enhancing the drone's ability to operate safely indoors.

- Validate the sensor measurements to ensure accuracy and reliability. This step involves testing the sensors in various indoor environments and calibrating them as needed.
6. **Integrate a companion computer (e.g., Raspberry Pi/ESP32) to implement obstacle avoidance algorithms.**
 - Use a companion computer to run advanced obstacle avoidance algorithms. The companion computer processes sensor data in real-time and sends commands to the flight controller to avoid obstacles, ensuring safe navigation in complex indoor environments.
 7. **Integrate a motion capture system for precise drone localization.**
 - Integrate with the existing motion capture system to provide accurate and real-time localization data for the drone.